

Compress the Response, Not the Data

Three everyday pictures of response-preserving aggregation

The one idea

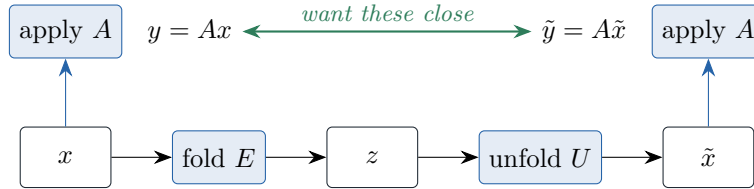
Every compression has the same shape: take data x , fold it down to a small code z , unfold it back to an approximation $\tilde{x}$.

$$x \xrightarrow{\text{fold } E} z \xrightarrow{\text{unfold } U} \tilde{x}.$$

The usual goal is $x \approx \tilde{x}$. But often we do not care about x itself — we care about what a *system* does to it. If a matrix A turns input into the thing we actually observe, the real goal is

$$\boxed{Ax \approx A\tilde{x}} \iff A(I - UE)x \approx 0.$$

Two inputs may be far apart yet produce the same output; two may be close yet produce different outputs. **Merge by response, not by appearance.** That single sentence is the whole method. The traffic version is identical: do not insist the OD table d be reconstructed; insist that the link flow $v = Hd$ be preserved, where $H = B^\top R$ is the OD-to-link response.

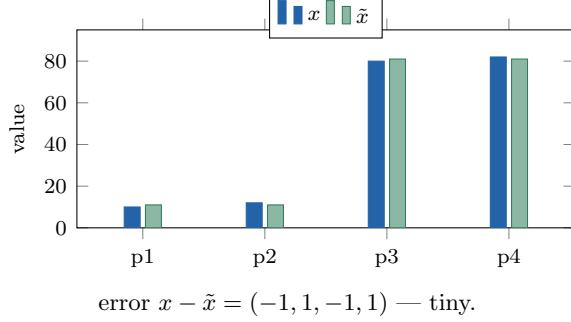


Example 1 — Image compression: merge by response

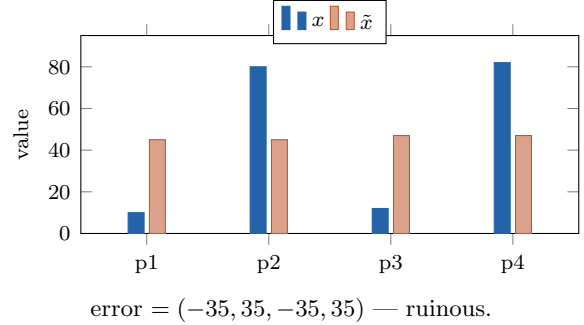
A tiny grayscale image, four pixels. Group them into two super-pixels by averaging (the fold E), then paint each group back with its average (the unfold U):

$$E = \frac{1}{2} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}.$$

Good grouping.
 $x = (10, 12, 80, 82)$ — adjacent pixels already alike.



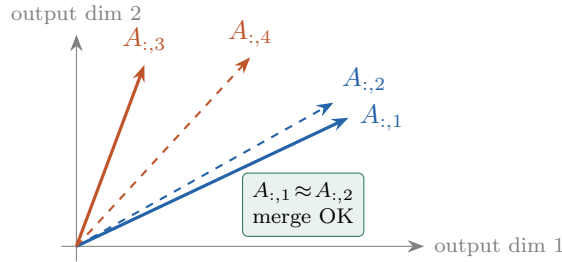
Bad grouping.
 $x = (10, 80, 12, 82)$ — same fold, but it mixes dark with bright.



The fold matrix is the same in both panels. The only difference is *which* pixels got merged. **Lesson:** aggregate things that respond alike, not things that sit next to each other.

Example 2 — Equalizer: preserve what is heard

Four frequency-band volumes feed a speaker: $x = (\text{bass}_1, \text{bass}_2, \text{mid}, \text{treble})$. Nobody hears x . What reaches the ear is the speaker response $y = Ax$, where column $A_{:,k}$ is how band k moves the output. If two bands push the output the same way, $A_{:,1} \approx A_{:,2}$, they can be merged into one super-band with no audible loss — even though they are different inputs.



The target is not $x \approx \tilde{x}$ but $Ax \approx A\tilde{x}$, i.e. $A(I - U)E)x \approx 0$. This is exactly the traffic statement, term for term:

Equalizer	Traffic
x : band inputs	d : OD demand
A : speaker response	$H = B^T R$: OD-to-link response
$y = Ax$: sound heard	$v = Hd$: link flow
merge similar bands	merge response-similar zones
keep the sound undistorted	keep corridor flow undistorted

Example 3 — Student grades: the low-rank code behind SVD

A student $\times$ course score matrix. The first two courses behave like math/physics, the last two like literature/history:

$$X = \begin{bmatrix} 90 & 92 & 30 & 28 \\ 88 & 91 & 32 & 31 \\ 25 & 27 & 85 & 88 \\ 28 & 30 & 82 & 86 \end{bmatrix}.$$

Four columns, but really only two abilities drive everything: a STEM score and a humanities score. The effective rank is 2, not 4.

	courses				
	90	92	30	28	
	88	91	32	31	STEM block
students	25	27	85	88	humanities block
	28	30	82	86	

The SVD reads off those two abilities automatically:

$$X \approx \underbrace{U_2 \Sigma_2}_{\text{decoder}} \underbrace{V_2^\top}_{\text{encoder}} .$$

$V_2^\top$ encodes four course scores into two latent ability scores; $U_2 \Sigma_2$ decodes them back into the grade pattern. SVD is the *optimal* low-rank fold/unfold — but its two “abilities” are abstract mixtures, not nameable courses.

Bridge: from these pictures to super-zones

Super-zone aggregation is the *interpretable* cousin of SVD. The fold is the OD aggregation $E = Q^\top = (S \otimes S)^\top$ and the unfold is U , so the compressed demand is $\tilde{d} = U Q^\top d$.¹ Both are fold/unfold pipelines:

$$\text{SVD: } H \approx U_s \Sigma_s V_s^\top \quad \text{vs.} \quad \text{super-zone: } d \rightarrow Q^\top d \rightarrow U Q^\top d.$$

We choose the zone map S so the interpretable fold tracks the optimal one on the corridors that matter:

$$H U Q^\top \approx U_s \Sigma_s V_s^\top \iff H d \approx H U Q^\top d \text{ on the dominant corridor modes.}$$

SVD says how many modes exist (the rank floor); the super-zones give names to them (geographic groups) while preserving the same dominant response.

The talking script. Image compression shows that fold/unfold is just merging similar items and expanding them back. The equalizer shows the thing to preserve is not the input but the system response Ax . SVD gives the optimal low-rank fold/unfold and tells you how many modes are real. Traffic demand compression is the same logic with names attached: $Q^\top$ folds the OD table, U unfolds it, $H = B^\top R$ is the system response. We never demand that the OD table be preserved — only that Hd and $HUQ^\top d$ agree on the leading low-rank corridor modes.

¹The unfolding operator U and the SVD left factor U_s are different objects; they share a letter only by convention.